Free Tier Friendly Three-Tier E-Commerce Architecture

Internet and Cloudflare Layer

Internet Users

Cloudflare DNS (External DNS)

DNS for app domain

Cloudflare Proxy/CDN/DDoS Protection

AWS Global Services

Amazon CloudFront (CDN/Cache)

Static

Amazon S3 (Static Assets)

Encryption where supported

Administrator

IAM (Users, Roles, Policies, MFA)

AWS API activity

Dynamic

Systems Manager Session Manager

Security

Amazon CloudWatch (Metrics, Logs, Dashboards)

CloudWatch Alarms

Amazon SNS (Email Notifications)

SSM Parameter Store (App config, non-secrets)

AWS CloudTrail (Mgmt Events)

AWS KMS (where applicable)

AWS Budgets (Free Tier alerts)

Cost Explorer

Inbound rules

Replication/Backup

Secondary AWS Region — CONCEPTUAL – NOT DEPLOYED

AMI create

AMI create

AMI Copies

Snapshot copy

Snapshot copy

EBS Snapshot Copies

S3 Replicated/Backup Data

Restore/Copy

RDS Recovery (Point-in-time)

Conceptual only — no resources created

Email

Production VPC (CIDR 10.0.0.0/16)

Presentation Tier

Application Tier

Database Tier

Monitoring

Internet Gateway

EC2 Auto Scaling Group (Linux Web)

min 1 / desired 1 / max 2

EBS Volumes (Web)

EC2 Backend (Linux)

EBS Volumes (App)

Multi-AZ + Prod enhancement (not deployed)

Info

Amazon RDS MySQL (Private, no public access)

RDS Automated Backups

CloudWatch Logs

Applied at ENI

Allowed traffic

Applied at ENI

Allowed traffic

Applied at ENI

Allowed traffic

Security Groups

Allowed traffic

No NAT Gateway (Private egress avoided)

Network ACLs

Development Workflow

Developers

Git

GitHub (Source + Docs)

Simple Git-based deploy (Linux shell to EC2)

Cost Management

Trusted Advisor (Free checks)

Resource Tags

Resource Groups

AWS Health Dashboard

Legend

IMPLEMENTED – Free Tier Friendly

OPTIONAL – Only within Free Allowance

CONCEPTUAL – Production/DR Design